

Sofiia Tretiak

AI and Python Engineer | Medical Informatics

tretiak.sofiia03@gmail.com · Vienna, Austria · [linkedin.com/in/sofia-tretiak](https://www.linkedin.com/in/sofia-tretiak) · github.com/TSofi

Profile

Medical informatics engineer working with LLMs, machine learning and clinical data. Built and open-sourced PharmaRAG, a multilingual RAG assistant that answers medicines questions from official drug registers with cited sources. Trained GRU-Transformer models on EEG data at a neuroengineering lab, and spent a year at Volvo turning user requests into engineering tickets and validating production data with SQL. MSc Medical Engineering and eHealth, Technikum Wien. Ukrainian native, English and Polish C1, German A2.

Projects

PharmaRAG - multilingual medicines assistant | personal project, live and open source | pharma-rag-nine.vercel.app

- Built and shipped the whole system alone: ingestion from six national drug registers, cleaning and normalisation, multilingual embeddings and ~400k chunks in a Qdrant vector database, retrieval with hybrid re-ranking
- LLM layer with Groq and a Gemini fallback: answers with source citations, safety tiers for medical content, six answer languages
- FastAPI on Render, React on Vercel, 36 automated tests, answers in 2-4 seconds; live and open source

Technologies: Python, FastAPI, Qdrant, embeddings, Groq and Gemini APIs, React, Render, Vercel, pytest

GEN-DOCK - molecular docking pipeline with clinical data mapping | MSc project, FH Technikum Wien

- Containerised docking pipeline: receptor and ligand preparation, Vina runs, scripted parsing of binding affinities into structured data, results mapped to HL7 FHIR resources for a clinical decision support view

Technologies: Python, AutoDock Vina, RCSB PDB, PubChem, PyMOL, Docker, HL7 FHIR R4

EyeSpy - Real-Time Emotion and Fatigue Detection | Solvro Summer Challenge 2025

- ResNet18 on FER-2013 with per-class failure analysis, integrated into a real-time video pipeline with MediaPipe landmark tracking (EAR/MAR fatigue metrics)

Technologies: Python, PyTorch, ResNet18, OpenCV, MediaPipe

Career Shark - Full-Stack Web Application | BEST Hackathon, Top 10

- Team lead of four; 7th out of about 50 teams with a full-stack product (React, FastAPI, MongoDB) built and deployed in 24 hours

Technologies: React, FastAPI, Python, MongoDB

Experience

Research Trainee, Neuroengineering Department

METAM, Karadeniz Technical University | Trabzon, Türkiye | Jun 2026 – Aug 2026

- Designed, trained and evaluated GRU-Transformer models in PyTorch for 9-class emotion recognition from multichannel EEG
- Engineered differential-entropy and functional-connectivity features; ran ablation studies on data volume, channels and signal duration

AI Output Reviewer and Data Annotator

Outlier and Alignerr | Remote | Mar 2026 – Present

- Evaluate LLM outputs across text, code and structured data against detailed rubrics: hallucinations, edge cases and systematic failure modes

Software Engineer Intern, Digital Product Team

Volvo | Wrocław, Poland | May 2025 – May 2026

- Clarified what users actually needed and turned it into clear tickets for engineering; handled 5-15 requests a day from first to third level in ServiceNow and Azure DevOps
- Ran SQL queries and prepared views against production databases to validate data and confirm root causes; tested REST API integrations
- Worked on a company-wide migration of user and access management across several countries; documented it and presented progress to stakeholders

Skills

AI and LLM: RAG architectures, embeddings and vector databases (Qdrant), LLM API integration (Groq, Gemini, Claude), prompt design, retrieval and output evaluation, AI-assisted development

Machine Learning: PyTorch, Transformer and GRU models, CNNs (ResNet), feature engineering, ablation studies, evaluation protocols, scikit-learn, OpenCV, MediaPipe

Engineering: Python, FastAPI, SQL, React, Docker, Git, REST APIs, JSON, pytest, cloud deployment (Render, Vercel), Azure DevOps, Jira

Analysis and Delivery: requirements clarification with non-technical stakeholders, data validation and reporting, process and system documentation, test cases, working between business and engineering teams

Domain: medical informatics, HL7 FHIR, clinical and regulatory data, bioinformatics (molecular docking, PDB structures), biomedical signal processing

Education & Certifications

Master of Science, Medical Engineering and eHealth

UAS Technikum Wien, Vienna | 2026 – 2028

Bachelor of Engineering, Medical Informatics and Bioengineering

Wrocław University of Science and Technology | 2022 – 2026

Languages

Ukrainian (native), English (C1), Polish (C1), German (A2, actively learning)